

Curriculum Vitae

Dante A. Capone

dcapone@ucsd.edu | 707-326-8161 | GitHub: github.com/DanteCapone | ORCID: 0009-0005-1232-359X

Updated December 2025

Education

Ph.D. in Oceanography with a Specialization in Interdisciplinary Environmental Research

University of California, San Diego Scripps Institution of Oceanography

- **Advisor:** Dr. Moira Décima
- **Thesis Committee:** Moira Décima, Andrew Barton, Michael Landry, Jules Jaffe, Jonathan Shurin, Lisa Cartwright
- **Title:** “Plankton Community Structure in the California Current Ecosystem Across Environmental Gradients and Through Episodic Extremes”
- Impacts of the 2020 lightning complex fires on phytoplankton dynamics in the Monterey Bay. Growth and grazing dilution and incubation field experiments assessing the impact of wildfire ash on plankton community dynamics. Rapid response oceanographic sampling of water biogeochemistry and aerosols during LA 2025 wildfires.
- Collection, processing, and analysis of patterns in zooplankton community composition using DNA metabarcoding and Zooscan imaging across environmental gradient in the California Current Ecosystem. Application of novel experiment and model to mitigate PCR bias in metabarcoding samples.
- Interdisciplinary work with visual artists both integrating humanities into oceanography and leveraging visual arts/humanities methods to critically understand the role of visual data in oceanography.

M.Sc. Marine Biology

University of California, San Diego Scripps Institution of Oceanography, 2022

B.Sc. Marine Biology *summa cum laude*

University of California, Santa Cruz, 2015-2019

B.Sc. Earth Science – Ocean Science Concentration *summa cum laude*

University of California, Santa Cruz, 2015-2019

- Thesis: Adviser: Dr. Matthew Clapham; “Climate or Cetaceans: Uncoiling the Mystery of Cenozoic Nautiloid Extinction”; Used optimal linear estimation to determine likely extinction times for Cenozoic nautiloids and hypothesize drivers.

Research Skills

Lab	Field	Data Analysis
· Chlorophyll-a fluorometry, flow cytometry, DNA extraction, metabarcoding	· Oceanographic sampling: CTD deployment and water sampling, zooplankton nets (Bongo, MOCNESS, ring net, IKMT), UVP6, aerosol samplers, sediment coring	· MATLAB, R, Python, GitHub
· Zooscan analysis		· Highly experienced at data wrangling, processing, analysis and manipulation.
· Isotope and biomass sample preparation	· Dilution/incubation experiments	· Bayesian modeling (<i>fido</i> , optimal linear estimation)
· Microscope sorting		· Time series analysis
		· Machine/Deep learning
		· Coupled ocean modeling

Publications (published, submitted and in prep)

Capone, D., Daniel, P., Kudela, R., Barton, A.D., Kahru, M. and Décima, M., 2025. Extreme wildfire conditions shift coastal phytoplankton community structure in California. *Limnology and Oceanography*.

Capone, D.A., Budroe, H., Gallego, R., Matthews, S., Nixon, M., Bowman, J., Silverman, J., Décima, M. Quantifying and correcting PCR bias in marine zooplankton metabarcoding using variable cycle calibration. *Molecular Ecology Resources*, submitted.

- Capone, D.A.**, Budroe, H., Gallego, R., Matthews, S., Nixon, M., Bowman, J., Silverman, J., Décima, M. Drivers of Zooplankton Community Structure and Diversity Across an Upwelling Gradient in the California Current Ecosystem. *Molecular Ecology*, in prep.
- Freilich M.A., Gangrade, S., Beaudin, E., Luis, K., Leyba, I., Omand, M., Lang, S., O' Neill, L., **Capone, D.A.** Open ocean biogeochemical impacts of extreme precipitation. submitted to *PNAS brief reports*.
- Erazo, N., Light, T., **Capone, D.A.**, Effinger, A., Erazo, P., Huang, L., Kannad, A., Lanpher, K., Norris, E., Perry, S., Romero, E. 2023. The Ocean as a Classroom: Considering the roles of equity, diversity, and justice in oceanographic knowledge production to promote accessibility for future generations. *Oceanography*. <https://doi.org/10.5670/oceanog.2024.109>
- Ashford, O.S., Guan, S., **Capone, D.**, Rigney, K., Rowley, K., Orphan, V., Mullin, S.W., Dawson, K.S., Cortés, J., Rouse, G.W. and Mendoza, G.F., 2021. A chemosynthetic ecotone—"chemotone"—in the sediments surrounding deep-sea methane seeps. *Limnology and Oceanography*, 66(5), pp.1687-1702. <https://doi.org/10.1002/lno.11713>
- Ashford, O.S., Guan, S., **Capone, D.**, Rigney, K., Rowley, K., Cordes, E.E., Cortés, J., Rouse, G.W., Mendoza, G.F., Sweetman, A.K. and Levin, L.A., 2021. Relationships between biodiversity and ecosystem functioning proxies strengthen when approaching chemosynthetic deep-sea methane seeps. *Proceedings of the Royal Society B*, 288(1957), p.20210950. <https://doi.org/10.1098/rspb.2021.0950>
- White, M., James, W. R., Lemoine, N. P., Peters, J. R., Stier, A. C., Emery, K. A., Castorani, M. C. N., **Capone, D. A.**, Štajner, A., Enright, L. N., Grier, S. R., Cawley, G. F., Spivak, A. C., Nelson, J. A., Hopcroft, R. R., Chen, A., Lyon, N. J., Kui, L., Caselle, J. E., Strickland, B. A., Allgeier, J. E., Rehage, J. S., and Burkepile, D. E. n.d. "Scale-Dependent Effects of Species Richness and Asynchrony Regulate the Temporal Stability of Consumer-Mediated Nutrient Dynamics." submitted for publication in *Proceedings of the National Academy of Sciences*.

Fellowships, Funding, Awards

NSF Graduate Research Fellowship Program, 2022–present
 UCSD SEED Funding (Wildfire rapid response sampling) 2022-2025
 Program for Interdisciplinary Environmental Research Fellowship (2020-2021)
 Regents Fellowship (2020-2021)
 2020 OSM Travel Award
 2019-2020 NSF GRFP Honorable Mention
 2015-2019 UCSC Regents Fellowship
 2015-2019 Dean's List Honors
 2015-2019 UCSC Scholar Athlete
 2019 UCSC Earth and Planetary Science Outstanding Senior
 2019 ASLO Multicultural Program Award
 2018-2019 UCSC NCAA Highest Athlete GPA

Cruise Experience

LA Fire Rapid Response Cruises. LA and Orange County Sanitation. 1/2025-present. Rapid response collection of aerosols and water samples for trace metals and biogeochemistry during and following LA Fires. Multiple 1-3 day cruises in nearshore regions in collaboration with sanitation districts.

SR2502 Santa Lucia Escarpment Cruise. UC Ship Funds. 2/2025. Ecological survey of Santa Lucia Escarpment. Assistant lead of zooplankton net operations and processing. Interdisciplinary collaboration with humanities scholars.

SR2422 Ancient California Ecosystems & Shorelines (ACES) Cruise. UC Ship Funds. *R/V Sally Ride*. Deployment and processing of sediment cores (gravity, box, Kaston) and clean room operations for ancient DNA. Opportunistic wildfire sampling.

RV Sproul Biological Oceanography Class Cruise. 11/2024. Bongo net deployments.

SR2307 Sub-mesoscale Ocean Dynamics Experiment (S-MODE) IOP2. NASA. *R/V Sally Ride*. 4/2023-5/2023. Dilution growth and grazing experiments, CTD casts, aerosol sampling coordinated with aerial and autonomous measurements of fast-moving features.

SR2215 Student Course Cruise. 12/2022. Cruise to expose students to oceanographic methods including Bongo nets, aerosol and CTD sampling. *Did not sail due to positive COVID test.

RV Sproul Deep-Sea Biology Cruise, 3/2022. IKMT, Otter trawl, sediment core sampling for course.

RV Sproul Moving-Vessel Profiler Cruise. UC Ship Funds. 12/2021. Deployed Moving Vessel Profiler imaging system and conducted oblique ring net tows

SP2122 Jaffe Cruise, 10/2021. Interdisciplinary/educational cruise: engaged with and taught visual artists and exhibit designers working on exhibits for the Birch Aquarium and The Getty Museum.

P2107 CCE-LTER Process Cruise. CCE-LTER. *R/V Roger Revelle*. 7-8/2021. Procedural deployments and processing of oblique Bongo, MOCNESS, and Salp net tows; trace-metal clean aerosol sampling; collaborative dilution incubations testing ash impacts on growth/grazing; grow-out experiments testing bioavailability of Fe of fecal pellets collected from *Euphausiia pacifica* and *Pyrosoma atlanticum*

SR2007 SiphWeb Cruise. *R/V Sally Ride*. 8-9/2020. Deployments and processing of 1m² and 10m² MOCNESS nets; CTD water sampling.

Additional Collaborative Projects

- **Consumer Functional Diversity LTER Synthesis Working Group (2025-present)**. Aquatic system synthesis exploring functional diversity of consumers, relationship to ecosystem function, and influence of temperature.
- **Consumer Mediated Nutrient Dynamics LTER Synthesis Working Group (2023-present)**. Synthesis of drivers of consumer nutrients across 10 marine LTER sites and response to disturbance.
- **NASA S-MODE Campaign (2023-present)**. Contribution of growth and grazing rate dilution experiments and UVP6 particle flux measurements to investigate the role of top down processes in phytoplankton dynamics at the submesoscale.
- **Ecological Investigation of the Santa Lucia Escarpment (2022-present)**, UC Ship Funds Student project. Contributions to zooplankton, phytoplankton, and biogeochemical data collection and analysis and interdisciplinary collaboration with humanities researchers.
- **Blue Humanities Anthology**. Interdisciplinary, collaborative work with UCSD visual arts researchers exploring our shifting relationship with oceanographic research in an increasingly digitized ocean.

Pre-Graduate Research Experience

Coastal Oceanographic Data Analyst – Bodega Marine Lab (2019-2020)

Advisor: Dr. John Largier. Analyzed California Coast surface current data from high-frequency radar measurements. Extensive spatio-temporal assessment of low-pH, low dissolved oxygen intrusions into San Francisco Bay using multi-sourced oceanographic data.

UCEAP Marine Biology/Terrestrial Ecology Student – University of Queensland (Fall 2019).

Designed, proposed, conducted, and presented collaborative research projects at Moreton Bay Research Station and Heron Island Marine Station and communicated results to local community and cohort through production of an infographic and an oral presentation.

Volunteer– University of Queensland Pandolfi Lab (Fall 2019).

Prepared and tested coral samples for nutrient content analysis to determine differences in nutrient uptake and growth between endemic and ubiquitous species between tropical and subtropical sites

Lab Volunteer– UCSC Palkovacs Freshwater Ecology Lab.

Identification and counting of zooplankton from Sierra Lake mesocosms; chlorophyll-A measurements.

Scripps Undergraduate Research Fellow REU– Scripps Institution of Oceanography (Summer 2018)

Advisors: Dr. Lisa Levin & Dr. Oliver Ashford. Measured and modeled ecosystem functioning of methane seep deep-sea macrofauna.

Lab Assistant – Ravelo Paleooceanography Lab (2017-2018).

Preparation and processing of sediments for stable isotope and Mg/Ca analysis; picking and identifying of benthic and pelagic foraminifera species.

Lab Assistant –UCSC Stable Isotope Lab (Summer 2017).

Preparation and packaging of samples for stable isotope analysis; Data entry and management.

Lab Volunteer – Bodega Marine Lab (Summer 2017).

Small boat estuarine field sampling (CTD casts, water sampling), data analysis in MATLAB.

Intern-UCSC Puma Project (2017).

Photo-tagging and analysis using Picasa 3; tracking and mapping mother puma data using Google Earth; literature review.

Professional Service and Communication

- Reviewer, *Limnology and Oceanography*, *Global Biogeochemical Cycles*

- Graduate Writing Fellow, Long Term Ecological Research Network Office, 2023-2024
- Co-founder, *in situ* – Scripps Institution of Oceanography Community Zine, 2024-present
- Scripps Art Committee, 2024-present
- Scripps Graduate Student Council BO Rep, 2023-2024
- Scripps Graduate Student Council Co-Chair, 2021-2023
- UCSD Graduate and Professional Student Association Rep, 2020-2022
- NCAA Track and Cross Country Captain 2016-2019
- Student Athletic Advisory Committee Representative 2015-2019

Outreach

- 2025 LA Wildfire press interviews and panel (KTLA, Bloomberg, Surflife, Ocean Science Trust, etc.)
- Scripps Community Outreach for Public Education (SCOPE) volunteer (2023-present)
- Stay Cool for Grandkids Teacher (2025-present)
- Birch Aquarium Summer Camp Sharing Science Activity Lead (2024, 2025)
- Bridge Builders San Diego Volunteer (2023, 2024)
- Birch Aquarium Happy Hour Speaker (2023)
- ASLO/OSM Storytellers (2019, 2020)

Mentorship & Teaching

- SIOC210 TA, Fall 2025
- CCE-LTER REU co-mentor 2025 for Nichole Ruiz. Project: Satellite and data analysis of LA Fire impacts on ocean.
- NSF GRFP Working Group Lead, 2023-present
- Undergraduate Research Scholar mentor, 2024
- CCE-LTER REU mentor, 2022 for Hannah Budroe. Project: DNA metabarcoding of marine zooplankton in the California Current Ecosystem
- Interdisciplinary Forum on Environmental Research Organizer, 2021-2022
- San Diego Tutor Tree volunteer, 2020-2021
- EART5 TA, 2018

Invited Talks and Lectures

- Scripps Ecology Seminar, June 2025, La Jolla, CA. Oral presentation
- VIS, May 2025, “Visualizations in Oceanography”
- SIO90, Fall 2025, “Ashes to Algae: Impacts of Wildfires on the Ocean”.

Posters and Presentations

- Ocean Sciences Meeting, 2/2026, Glasgow, Scotland. Oral presentation
- Scripps Ecology Seminar, June 2025, La Jolla, CA. Oral presentation: “Patterns of Zooplankton Diversity Across the California Current Ecosystem”
- RUSTED/Iron at the Air-Sea Interaction Workshop, July 2025, Asheville, NC. Oral presentation
- Aquatic Sciences Meeting, 6/2025, Charlotte, NC. Oral presentation
- Ocean Carbon Biogeochemistry Summer Workshop, 6/2024, Woodshole, MA. Oral presentation
- California Ocean Observing Meeting, 5/2024, San Diego, CA. Poster
- NASA S-MODE Meeting, 11/2023, Mountain View, CA. None.
- CCE Forum, 9/2023, La Jolla, CA. Oral presentation
- LTER All Scientists Meeting, Pacific Grove, CA. None.
- Aquatic Sciences Meeting, 2019, San Juan, Puerto Rico. Poster: “Are Methane Seep Sediments a Hotspot of Ecosystem Functioning in the Deep Sea”

Workshops and Meetings

- RUSTED/Iron at the Air-Sea Interaction Workshop, July 2025, Asheville, NC. Oral presentation
- Aquatic Sciences Meeting, 6/2025, Charlotte, NC. Oral presentation
- CCE Forum 2024, La Jolla, CA. Oral presentation
- Ocean Carbon Biogeochemistry Summer Workshop, 6/2024, Woodshole, MA. Poster:
- California Ocean Observing Meeting, 5/2024, San Diego, CA. Poster

- NASA S-MODE Meeting, 11/2023, Mountain View, CA. None.
- CCE Forum, 9/2023, La Jolla, CA. Oral presentation: “
- LTER All Scientists Meeting, 9/2022, Pacific Grove, CA. None.
- Aquatic Sciences Meeting, 2019, San Juan, Puerto Rico. Poster: “Are Methane Seep Sediments a Hotspot of Ecosystem Functioning in the Deep Sea”

Relevant Coursework

		<u>Discussions/Seminars</u>
· Biological Oceanography	· Shipboard Knowledge	· California Current Ecosystem
· Physical Oceanography	Production	Dynamics
· Marine Chemistry	· Science Communication	· Marine LTER Sites
· Analysis of Physical	· Professional Ethics	· Ecological Genomics of
Oceanographic Data I & II	· Design Communication	Marine Protists
· Deep Learning	· Interdisciplinary Forum	· Multiple Stressors
· Marine Zooplankton	· Geophysical Fluid Dynamics	· Ecology Seminar
· Complex Systems Analysis	(UCSC)	Presentations
· Deep Sea Biology	· Ocean Modeling (UCSC)	
· Pelagic Ecology	· Freshwater Ecology (UCSC)	

Extracurricular Activities

- NAUI open water certification (2018)
- US Boater Certification/CA Boater Card (2021)
- Semi-Elite marathon runner for Lane Zero Athletics Club.
- Science-inspired Illustration and T-shirt designer